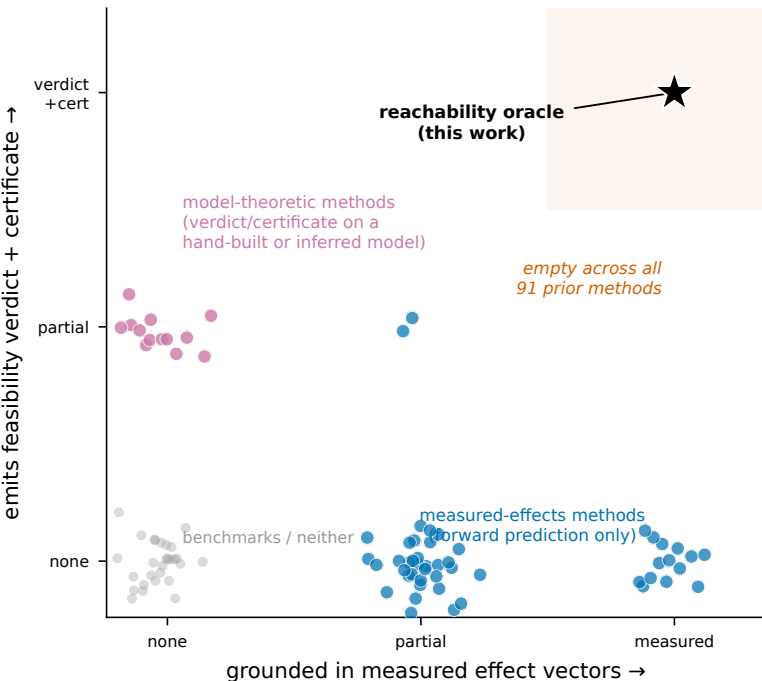


a No prior method is grounded in measured effects and emits a verdict + certificate (n=92 surveyed)



b Modality triage over 102 real nominations

45/102 (44%) hard-to-drug · 10 of 102 have a clinical-grade drug

GO target reachable

→ minimal ranked knockdown set → arrayed screen

green-light: JAK2, ICOS, MAPK14, CD3D

REDIRECT activation-required

→ switch to CRISPRa / agonist
→ certificate names genes to test

tractable-but-untried: IL7R, ZAP70, TET2

STOP provably outside the cone

→ do NOT spend a CRISPRi arm on it

required-but-undruggable: IRF1

all nominations are wet-lab hypotheses, not validated targets